

Probability theory

Guo Haoyang

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1 Overview

Analysis	Probability
Measure space (X, Σ, μ)	Sample space $(\Omega, \mathcal{F}, \mathbb{P})$
measure μ	probability measure $\mathbb{P}$
measurable sets $A \in \Sigma$	events $A \in \mathcal{F}$
measurable functions $f : (X, \Sigma, \mu) \rightarrow (Y, \Xi, \lambda)$	random variables $X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (E, \mathcal{A})$
pullback σ -algebra by f	σ -algebra generated by X
Lebesgue integration of $f, \int f d\mu$	expectation of $X, \mathbb{E}[X]$
Radon-Nikodym derivative w.r.t. two measures, $\frac{d\mu}{d\lambda}$	cumulative distribution function $F(x) = \mathbb{P}(X \leq x)$
L^p	probability density function, probability mass function
convergence in measure	having p -th moment
almost everywhere, a.e.	convergence in probability (measure)
characteristic function	almost surely, a.s.
Fourier transformation of a measure	indicator function
	characteristic function of a distribution

Table 1: Real Analysis and Probability Theory

In the following sections we want to establish the probability theory on the arbitrary and the most abstract measurable space rather than $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Everything will be set up on the abstract measurable space $(E, \mathcal{A})$, unless we stress the case relying on $\mathbb{R}$.

2 Fundamental settings for probability theory

Definition 2.1: Random variables

A random variable X is a measurable function on a probability space.

Example There are some special random variables: real-valued random variables and random graphs. The real-valued random variables are the center of our study in probability theory.

Definition 2.2: Pushforward measure by a random variable

Given a random variable defined on the probability space to a measurable space $(E, \mathcal{A})$, the **pushforward measure** of $\mathbb{P}$ under X , denoted by $X_*\mathbb{P}$, is defined as: $\forall A \in \mathcal{A}$,

$$X_*\mathbb{P}(A) := \mathbb{P}(X^{-1}(A))$$

Proposition 2.1: Pushforward measure is a measure

Let all the settings be the same as in 2.2. The pushforward measure defined above is a measure.

Proof:

□

2.1 Distribution function

2.1.1 Example: Cumulative distribution function (CDF)

Our study of the distribution function will cross the following steps as shown in the following diagram:

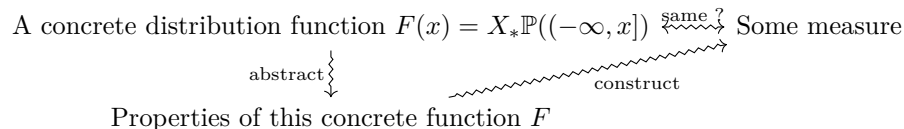


Figure 1: Main threads for the study of cumulative distribution functions

First, we start with a concrete cumulative distribution function $F(x)$, which is uniquely determined by the pushforward measure $X_*\mathbb{P}$ of a random variable. We then abstract this function to study its core mathematical properties (non-decreasing, limits at infinity, right-continuous).

Next, we reverse the perspective: using the Lebesgue-Stieltjes theorem, we show that any abstract function satisfying these properties can be used to construct a unique, valid probability measure μ on the real line.¹

Finally, we close the loop (the 'same?' step) by proving that this newly constructed measure μ is identically equal to the original pushforward measure $X_*\mathbb{P}$. This establishes a perfect bijection or a characterization between valid CDFs and probability measures on $\mathbb{R}$.

Because the two measures are exactly the same, this diagram can be condensed into

$$\{\text{Abstract cumulative distribution functions } F\} \leftrightarrow \{\text{measures on } \mathbb{R}\}$$

Definition 2.3: Cumulative distribution function

Let $X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be a real-valued random variable. The cumulative distribution function with respect to X , denoted F , is

$$F(x) := X_*\mathbb{P}((-\infty, x])$$

Remark Usually this notation $X_*\mathbb{P}((-\infty, x])$ is written as $\mathbb{P}(X \leq x)$ in most textbooks.

Proposition 2.2: Properties of cumulative distribution functions

Let X be a real-valued random variable and let F be of the property $F(x) := X_*\mathbb{P}((-\infty, x])$. Then, F has the following properties:

- (1) F is non-decreasing.
- (2) (Limit to the right and limit to the left) $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.
- (3) (Right continuity) $\lim_{x \downarrow y} F(x) = F(y)$

We then extract these three properties and define an abstract distribution function f to have these three properties. The next question is if we are given an abstract distribution function, is it possible to reconstruct the measure, a Borel probability measure on the real line?

Definition 2.4: Abstract (cumulative) distribution function

Theorem 2.1: Correspondence between measures and distribution functions

The broad probability measures on $\mathbb{R}$ are in one-to-one correspondence with distribution functions. For each distribution function F , $\exists! \mathbb{P}$ so $\forall x \in \mathbb{R}$,

$$\mathbb{P}((-\infty, x]) = F(x)$$

¹This is the existence and uniqueness of the newly constructed measure μ . We haven't yet compared this new measure μ with the one $X_*\mathbb{P}$. We will do the comparison in the next step.

2.1.2 Example: Cumulative distribution functions for random vectors

Definition 2.5: Cumulative distribution function for random vectors

Let $\mathbf{X} : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ be a real valued multivariate random variable with components $X_1, \dots, X_n : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that

$$\mathbf{X} = (X_1, X_2, \dots, X_n)^\top$$

If $\mathbb{P}$ is a probability measure on $(\Omega, \mathcal{F})$, then $\mathbf{X}$ induced a probability measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$, $\mathbf{X}_*\mathbb{P}$. Then the **multivariate cumulative distribution** or the **joint probability distribution** of $\mathbf{X}$ is a function $F_{\mathbf{X}} : \mathbb{R}^n \rightarrow [0, 1]$ defined as

$$F_{\mathbf{X}}(\underline{x}) = \mathbf{X}_*\mathbb{P}\left((-\infty, x_1] \times \dots \times (-\infty, x_n]\right)$$

where $\underline{x} = (x_1, \dots, x_n)^\top$

Remark In an entrenched notation,

$$F_{\mathbf{X}}(\underline{x}) = \mathbb{P}\left(X_1 \leq x_1, \dots, X_n \leq x_n\right)$$

To demystify the notation $\mathbf{X}_*\mathbb{P}$, we try to expand it:

$$\begin{aligned} \mathbf{X}_*\mathbb{P}\left((-\infty, x_1] \times \dots \times (-\infty, x_n]\right) &= \mathbb{P}\left(\mathbf{X}^{-1}\left((-\infty, x_1] \times \dots \times (-\infty, x_n]\right)\right) \\ &= \mathbb{P}\left((X_1, X_2, \dots, X_n)^{-1}\left((-\infty, x_1] \times \dots \times (-\infty, x_n]\right)\right) \\ &\stackrel{\dagger}{=} \mathbb{P}\left(\bigcap_{i=1}^n X_i^{-1}((-\infty, x_i])\right) \end{aligned}$$

$\dagger$ is because:

$$\begin{aligned} &(X_1, X_2, \dots, X_n)^{-1}\left((-\infty, x_1] \times \dots \times (-\infty, x_n]\right) \\ &= \left\{\omega \in \Omega \mid (X_1(\omega), \dots, X_n(\omega)) \in (-\infty, x_1] \times \dots \times (-\infty, x_n]\right\} \\ &= \left\{\omega \in \Omega \mid X_1(\omega) \in (-\infty, x_1] \bigwedge \dots \bigwedge X_n(\omega) \in (-\infty, x_n]\right\} \quad \text{definition of Cartesian prod} \\ &= \left\{\omega \in \Omega \mid X_1(\omega) \in (-\infty, x_1]\right\} \cap \dots \cap \left\{\omega \in \Omega \mid X_n(\omega) \in (-\infty, x_n]\right\} \\ &= \bigcap_{i=1}^n \left\{\omega \in \Omega \mid X_i(\omega) \in (-\infty, x_i]\right\} \\ &= \bigcap_{i=1}^n X_i^{-1}((-\infty, x_i]) \end{aligned}$$

3 Integration and differentiation theory

3.1 Integration theory on probability

To talk about the integration theory on probability, we should first fix a random variable $X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (E, \mathcal{A})$. To do the integration we also need an integrand, i.e. a measurable function $f : (E, \mathcal{A}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$. The integration is always working on a real-valued measurable function.

For a given measure μ on the measurable space $(E, \mathcal{A})$, the integration with respect to μ is

$$\int_E f d\mu \tag{1}$$

On the other hand the probability measure $\mathbb{P}$ has a pushforward measure $X_*\mathbb{P}$ on the measurable space $(E, \mathcal{A})$. So we have another measure space $(E, \mathcal{A}, X_*\mathbb{P})$. This measure also gives an integration for f ,

$$\int_E f dX_*\mathbb{P} \quad (2)$$

In the next subsection we will study the relation between the two integrations (1) and (2).

3.1.1 Example: Expectation

Expectation is a concept that is only for a real-valued random variable.

Definition 3.1: Expectation

For a real-valued random variable $X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, the **expectation** of X , denoted $\mathbb{E}[X]$, is defined as

$$\mathbb{E}[X] = \int_{\Omega} X d\mathbb{P}$$

Definition 3.2: Variance

$$\text{Var}[X] := \mathbb{E}[(X - \mathbb{E}[X])^2]$$

Proposition 3.1: Properties of Variance

Let X be a real-valued random variable and $c \in \mathbb{R}$.

- Var is additive, i.e. $\text{Var}[X + Y] = \text{Var}[X] + \text{Var}[Y]$.
- $\text{Var}[X + c] = \text{Var}[X]$
- $\text{Var}[cX] = c^2 \text{Var}[X]$

Proposition 3.2: Expectation of composition

Let $g : (\mathbb{R}, \mathcal{B}(\mathbb{R})) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be a Borel measurable function. Then,

3.1.2 Some applications and inequalities

Proposition 3.3: Markov's inequality (Probability)

Let X be a non-negative real value random variable and $a > 0$, then

$$X_*\mathbb{P}([a, +\infty)) \leq \frac{\mathbb{E}(X)}{a}$$

If adopting the notation $X_*\mathbb{P}([a, +\infty)) =: \mathbb{P}(X \geq a)$, then there is a rehash

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}(X)}{a}$$

3.2 Differentiation theory on probability

From the theorem 2.1, We know that every random variable with a broad probability measure must have a distribution function. So the distribution function always exists. The notions of density come from the cumulative distribution function, inspired by some famous distributions, like the uniform distribution, Gaussian distribution. In these cases, one can get some continuous functions $f : \mathbb{R} \rightarrow [0, +\infty)$, such that the integration of this continuous function is the distribution function. By the fundamental theorem of calculus, this f is the derivative of the given distribution function. But, it is not always the case that the so-called density function should be defined as a derivative of the distribution functions in the normal sense. Consider the following cases.

The Bernoulli distribution function is not continuous so it does not have a derivative globally.

Another two cases are pathological. The first is:

- Let $\{a_j\}_{j \in \mathbb{Z}}$ be a strictly decreasing positive sequence with the sum $\sum_{j=1}^{\infty} a_j = 1$ and let $\{q_j\}_{j \in \mathbb{Z}}$ be an enumeration of $\mathbb{Q}$. Consider the distribution function $F := \sum_{j=1}^{\infty} a_j \mathbb{1}_{[q_j, \infty)}$. This function is discontinuous on a dense set, so it is not differentiable.
- The counter devil staircase function: Let L_n be the left endpoint of each C_n receipt for a Cantor set C . Then consider the function F_n defined as the following:

$$F_n(x) = \begin{cases} 0 & x < 1 - a_{2^n} \\ \frac{k}{2^n} & 1 - a_{2^n - (k-1)} \leq x < 1 - a_{2^n - k} \\ 1 & 1 \leq x \end{cases}$$

Let F be the limit of this $\{F_n\}$, i.e. $F(x) := \lim_n F_n(x)$. Then this F is the so-called devil staircase function.

Still, let X be the random variable set above. Then, consider a measure μ defined on the measurable space $(E, \mathcal{A})$. To fix this disability when the density function is not a continuous function or when the distribution function F does not have a derivative, we use an extended idea of the derivative for density, the Radon-Nikodym derivative with respect to the pushforward measure $X_*\mathbb{P}$ and μ , $\frac{dX_*\mathbb{P}}{d\mu}$.

Definition 3.3: Density function

3.2.1 Examples: Probability density function and probability mass function (PDF & PMF)

When X is a real-valued random variable, this definition of the density function collapses to the derivative of the distribution function F .

Definition 3.4: Probability density function, probability mass function

When μ is the Lebesgue measure, the density (Radon-Nikodym derivative) is a probability density function.
When μ is a counting measure, the density (Radon-Nikodym derivative) is just a probability mass function.

3.3 Some common distribution and density functions on $\mathbb{R}$

In our following discussion, we have two types of distribution: discrete distributions and continuous distributions. First let's make sense of distributions. When we talk about distributions, we are talking about its random variables, its distribution function, and also its density function. Let $X : (\Omega, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ be a real-valued random variable. For the probability measurable space $(\Omega, \mathcal{F})$, assign it a probability measure $\mathbb{P}$. Talking about distributions and density functions entails doing integrations and differentiations on the Borel measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, implying that we need a measure on it.

Depending on the type of measures we assign to this space, we have two classes of distributions:

When $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is assigned by a Lebesgue measure λ , the distribution associated with this measure λ is called a continuous distribution.

When $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ assigned by a counting measure μ , the distribution associated with μ is known as a discrete distribution.

As we set, we shall first compare the measure λ or μ assigned to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ with the pushforward probability measure $X_*\mathbb{P}$. More precisely, we will first consider the Radon-Nikodym derivatives, $\frac{dX_*\mathbb{P}}{d\mu}$ or $\frac{dX_*\mathbb{P}}{d\lambda}$, between them. Then, we will do integrations to recover the CDF with respect to these Radon-Nikodym derivatives.

Even though we just mentioned some distributions of reputations, here we want to list all those different kinds of distributions together with their density functions and to study them as whole.

3.3.1 Discrete distributions

Binomial distribution

Poisson distribution

3.3.2 Continuous distributions

Uniform distribution

Gamma distribution

Normal/Gaussian distribution

Cauchy distribution

Exponential distribution

4 Independence

4.1 Different kinds of independence

The first definition is on the independence of two events. This scenario can be extended to the independence of finite events.

Definition 4.1: Independence of finite events

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Then events $A_1, A_2, \dots, A_n \in \mathcal{F}$ are **independent** $\Leftrightarrow$

$$\mathbb{P}\left(\bigcap_{i=1}^n A_i\right) = \prod_{i=1}^n \mathbb{P}(A_i)$$

For the independence of events, there are some generalizations of mutual independence:

$$\text{Mode of independence of events} \begin{cases} 2 \text{ rv s} & \text{pairwise} = \text{mutual} \\ \geq 2 \text{ rv s} & \begin{cases} \text{pairwise independent} \\ k\text{-wise independent} \\ \text{mutually independent} \end{cases} \end{cases}$$

We can do an abstraction to a higher layer or hierarchy. That is the dependence of two σ -algebras or the two subsets of σ -algebras that contain many events. This can be inductively generalized also to finite cases.

Definition 4.2: Independence of Sigma Algebras

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $\mathcal{F}_1, \mathcal{F}_2, \dots, \mathcal{F}_n \subseteq \mathcal{F}$ be σ -algebras. $\mathcal{F}_1, \mathcal{F}_2, \dots, \mathcal{F}_n$ are independent $\Leftrightarrow \forall i, \forall A_i \in \mathcal{F}_i,$

$$\mathbb{P}\left(\bigcap_{i=1}^n A_i\right) = \prod_{i=1}^n \mathbb{P}(A_i)$$

If we have some sigma algebras, we can use the random variables to pull back those σ -algebras. Then we can talk about the independence of random variables. The independence of random variables is actually built on the independence of the σ -algebras. The independence of random variables is extendable from two random variables to finite cases.

Definition 4.3: Independence of random variables

For a random variable $X : (\Omega, \mathcal{F}, \mathbb{P}) \rightarrow (E, \mathcal{A})$, this definition requires pulling back the whole σ -algebra $\mathcal{A}$, namely $X^{-1}(\mathcal{A})$. And then we consider the σ -algebra generated by this pullback, namely $\sigma(X^{-1}(\mathcal{A}))$. It's hard to check what this σ -algebra is. In fact, the structure of this σ -algebra is very simple. It is simply the pull-back $X^{-1}(\mathcal{A})$.

Proposition 4.1: The structure of σ -algebra generated by random variable

The independence among events $A_1, \dots, A_n$ could be rephrased in the language of random variables if we take indicator functions $\mathbb{1}_{A_1}, \dots, \mathbb{1}_{A_n}$ into considerations.

All these notions of independence are in the context of finite cases and all of these notions can be generalized to infinite cases. Any collections of infinite stuff are said to be independent if every finite sub-collection of those things is independent.

Practically speaking when we have a, say, an infinite list of sigma algebras, we want to check its independence so we need to pick every finite sub-collection of those lists of sigma algebras. It is very cumbersome and almost impossible to do this check. To get rid of this heavy work, we can instead do other things.

Theorem 4.1: $\pi - \lambda$ theorem

If $\mathcal{P}$ is a π -system and $\mathcal{L}$ is a lambda system that contains $\mathcal{P}$, then $\sigma(\mathcal{P}) \subseteq \mathcal{L}$.

Corollary 4.1: Independence from π -systems to σ -algebras

Suppose $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n$ are independent π -systems. Then, the σ -algebras generated by those π -systems, $\sigma(\mathcal{A}_1), \dots, \sigma(\mathcal{A}_n)$ are independent.

Proof: $\forall i$, let A_i be the set that is in $\mathcal{A}_i$, i.e. $A_i \subseteq \mathcal{A}_i$. Then, consider all the elements of $\mathcal{A}_1$ that are independent of the intersections of any other $\mathcal{A}_i$ s, namely

$$\mathcal{L} := \{A \in \mathcal{A}_1 : \mathbb{P}\left(A \cap \left(\bigcap_{i=2}^n A_i\right)\right) = \mathbb{P}(A) \cdot \mathbb{P}\left(\bigcap_{i=2}^n A_i\right)\}$$

From now on, let $H := \bigcap_{i=2}^n A_i$. *Claim:* $\mathcal{L}$ is a λ -system.

- $\Omega \in \mathcal{L}$, Because $\mathbb{P}\left(\Omega \cap \left(\bigcap_{i=2}^n A_i\right)\right) = \mathbb{P}(\Omega)\mathbb{P}\left(\bigcap_{i=2}^n A_i\right)$.
- If $A, B \subseteq \mathcal{L}$ with $A \subseteq B$, then $(B - A) \cap H = (B \cap H) - (A \cap H)$. Then,

$$\mathbb{P}((B - A) \cap H) = \mathbb{P}(B \cap H) - \mathbb{P}(A \cap H) \stackrel{\text{By def of } \mathcal{L}}{=} (\mathbb{P}(B) - \mathbb{P}(A)) \cdot \mathbb{P}(H) = \mathbb{P}(B - A)\mathbb{P}(H)$$

whence $B - A \in \mathcal{L}$.

- $\mathcal{L}$ is closed. Take a sequence $\{\mathcal{A}_n\}_n \subseteq \mathcal{L}$ such that $A_n \uparrow A$,

$$\mathbb{P}(A \cap H) = \lim_n \mathbb{P}(A_n \cap H) = \left(\lim_n \mathbb{P}(A_n)\right) \cdot \mathbb{P}(H) = \mathbb{P}(A)\mathbb{P}(H)$$

$\mathcal{A}_1$ is a σ -algebra, so it is a π -system. Applying the $\pi - \lambda$ theorem 4.1, it gives $\sigma(\mathcal{A}_1) \subseteq \mathcal{L}$. Notice that $\forall A_1 \in \sigma(\mathcal{A}_1)$, and $A_i \in \mathcal{A}_i$ for $2 \leq i \leq n$,

$$\mathbb{P}\left(\bigcap_{i=1}^n A_i\right) \stackrel{\text{By def of } \mathcal{L}}{=} \mathbb{P}(A_1)\mathbb{P}\left(\bigcap_{i=2}^n A_i\right) \stackrel{\text{by assumption}}{=} \prod_{i=1}^n \mathbb{P}(A_i)$$

The definition of independence of a collection of sets yields that $\sigma(\mathcal{A}_1), \mathcal{A}_2, \dots, \mathcal{A}_n$ are independent.

So far we have shown that: Suppose $\mathcal{A}_1, \mathcal{A}_2, \dots, \mathcal{A}_n$ are independent, then $\sigma(\mathcal{A}_1), \mathcal{A}_2, \dots, \mathcal{A}_n$ are independent.

Because independence is invariant under permutations, by applying this process to $\mathcal{A}_2, \dots, \mathcal{A}_n, \sigma(\mathcal{A}_1)$, we have $\sigma(\mathcal{A}_2), \dots, \mathcal{A}_n, \sigma(\mathcal{A}_1)$ are independent. Iterate this process n times, and then we have the desired result. $\square$

Why does this definition get rid of the heavy work of checking every finite subset? $A_n \uparrow A$

Up to now, we have dealt with independence among single random variables. The next step is to study the independence of the functions of disjoint collections of random variables. Imagine this case: Suppose we have four random variables X_1, X_2, X_3 , and X_4 . These random variables might not be mutually independent, but $\{X_1, X_2\}$ and $\{X_3, X_4\}$ are independent. If there is a measurable function on X_1 and X_2 , say $X := X_1 + e_2^X$, and there is also a measurable function on X_3 and X_4 , say $Y := X_3 \ln X_4$. So there are two random variables, X and Y . Are X and Y independent? The intuition is yes and the remaining part will rigorously work on this kind of problem.

In general, let's consider a square of random variables, $\{X_{ij}\}_{i,j}$ with $1 \leq i \leq n$ and $1 \leq j \leq m(i)$. Suppose for each i , the collections $\{X_{ij}\}_j$ are mutually independent. Assign a measurable function f_i for each collection. Because collections are disjoint, functions f_i when applied to the collections of random variables, is expected to be new random variables, denoted $Y_i := f_i(X_{i1}, \dots, X_{im(i)})$. Those Y_i s are expected to be mutually independent.

4.2 Independence on expectation and variance

Let $X_1, X_2, \dots, X_n$ be independent real-valued random variables. We study the correlations among those random variables.

Definition 4.4: Correlation

For variance, we have

Corollary 4.2: Independence for Variance

When $X_1, \dots, X_n$ are **independent** real-valued random variables such that $\mathbb{E}(X_i^2) < \infty$ and they are pairwise uncorrelated.

$$\text{Var} \left[\sum_{i=1}^n X_i \right] = \sum_{i=1}^n \text{Var}[X_i]$$

Proof: Let $S_n = \sum_i X_i$ and $\mu_i = \mathbb{E}[X_i]$. By the definition of variance:

$$\begin{aligned} \text{Var}(S_n) &= \mathbb{E}[(S_n - \mathbb{E}[S_n])^2] \\ &= \mathbb{E} \left[\left(\sum_i X_i - \sum_i \mathbb{E}[X_i] \right)^2 \right] \\ &= \mathbb{E} \left[\left(\sum_i (X_i - \mu_i) \right)^2 \right] \\ &= \mathbb{E} \left[\sum_i (X_i - \mu_i)^2 + \sum_{i \neq j} (X_i - \mu_i)(X_j - \mu_j) \right] \quad \left(\sum \alpha_i \right)^2 = \sum \alpha_i^2 + \sum_{i \neq j} \alpha_i \alpha_j \\ &= \sum_i \mathbb{E}[(X_i - \mu_i)^2] + \sum_{i \neq j} \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)] \\ &= \sum_i \text{Var}(X_i) + \sum_{i \neq j} 0 \quad X_i \text{ s are uncorrelated, } \forall i, j, \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)] = 0 \\ &= \sum_i \text{Var}(X_i) \end{aligned}$$

□

5 Convergence modes

In this section we will study Law of large numbers and the central limit theorem. We observe that each theorem corresponds to different modes of convergence. It might seem that these convergence theorems were formulated to serve the concepts of convergence. That gets the causality backwards. Convergence itself is not the ultimate objective. Instead, it is driven by the physical intuition and the real-world necessities, which can only be rigorously formulated through certain modes of convergence.

5.1 Modes of convergence

5.2 Law of Large Numbers

5.2.1 Weak Law of Large Numbers

5.2.2 Strong law of large numbers

Theorem 5.1: Strong law of large numbers

Let $X_1, X_2, \dots$ be an infinite sequence of i.i.d. Lebesgue integrable random variables with $\mathbb{E}[X_1] = \mathbb{E}[X_2] = \dots = \mu$, then let $\bar{X}_n := \frac{1}{n}(X_1 + X_2 + \dots + X_n)$

As $n \rightarrow +\infty$,

$$\bar{X}_n \rightarrow \mu$$

If $\text{Var}[X_1] = \text{Var}[X_2] = \dots = \sigma^2$ and no correlation between random variables, then

$$\text{Var}(\bar{X}_n) = \text{Var}\left(\frac{1}{n} \sum_{j=1}^n X_j\right) = \frac{1}{n^2} \text{Var}(X_1 + \dots + X_n) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$

Remark The strong law of large numbers dictates the fact that no matter how those random variables are distributed, when you consider their mean, the distribution of this mean random variable converges to a certain distribution.

Proof:

□

5.3 Central Limit Theorem

6 Conditional probability, Martingales